

Tom Moore's Web Page

(archived as of June 2014)



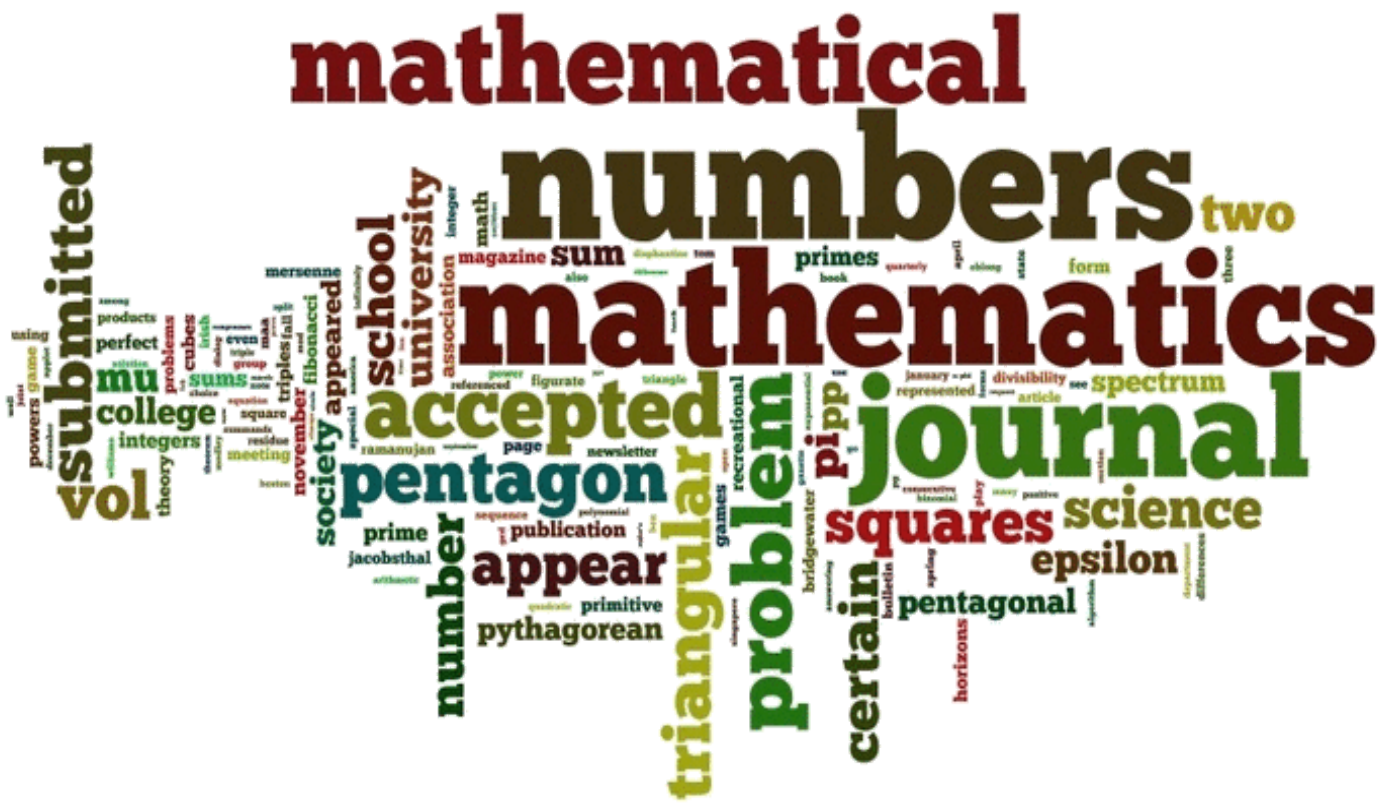
1944 ~ 2014

General Information

I am Professor Emeritus of Mathematics in the Department of Mathematics at [Bridgewater State University](#).

My degrees were obtained at [Stonehill College](#) and at [The University of Notre Dame](#), both schools run by [the Congregation of the Holy Cross](#).

I have retired as of December 31, 2012, after 44 years at BSU .



Mathematical Interests

My research interests are in finite group theory and number theory, as well as their connections to Recreational Mathematics. I also have an interest in Cryptography/Cryptology. See my publications below.

Pedagogical Interests

I am interested in the use of *technology* at all levels of the mathematics curriculum. For example in an introductory Linear Algebra course I have used MINimal MATlab from [Joel W. Robbin](#) which accompanies his 1995 *Matrix Algebra* textbook. I have most recently taught this course aided by MAPLE software from [Waterloo Maplesoft](#). I now use Maple 14 in various of my courses, especially Number Theory.

A great resource for software related to K-12 and collegiate level mathematics is [the Mathematics Archives WWW Server](#) at the University of Tennessee, Knoxville.

I am also interested in using *recreational mathematics* in my mathematics teaching, whenever possible, especially in Abstract Algebra courses. There, for example, I illustrate coset partitions using circular residue designs on the subgroups of the integers mod m or on the group of units mod m . We analyze peg-jumping games with the aid of the Klein four-group and sliding marker games on star graphs with the aid of Lagrange's Theorem. In this same spirit I have recently created a course and a manuscript called *Mathematical Games and Puzzles for K-12 Classrooms* (MATC 560).

My graduate student, Neil Roza, is in the process of creating Java Applets for some of the games from my manuscript. At this time you can play *Ducci's Game* which is a subtraction game. It consists of rounds of subtraction (which the applet does for you) starting with a quadruplet of integers of your choice and with the goal being to arrive at four zeroes in the greatest number of steps. Go [here](#) to play, answering "open" to the dialog box you will see displayed.

Also available is the *Pile Splitting Game*. A natural number N is used as input. Triple click on the starting default value of 1; type your starting value of N ; hit enter. The program generates the first split automatically into two summands. To change a summand just triple click on it and then type the value of your choice and hit enter. You then continue to split summands by double clicking on their folder symbols (and you can change the program's generated summands at any time with a triple click and by entering your choice). Continue splitting until you reach all 1s. For each split the applet will multiply the

summands and finally it will total all these products. A surprise awaits you as you play again with the same number N and split in different ways! Go [here](#) to play, answering "open" to the request in the dialog box you'll see.

Pretty pictures and more! Make your own *Residue Designs*. This applet illustrates the arithmetic/algebra in the group of reduced residues modulo N . You first specify a natural number N as modulus and a multiplier which must be less than N and relatively prime to N . Do this by clicking on the change numbers button. (The slow draw button option is currently not implemented.) The applet places all the reduced residues equally spaced around a circle. Then it multiplies each residue by your choice of multiplier. If the answer to multiplying x on the circle is y (reduced mod N) on the circle, then a chord is drawn joining x and y . Images can be very intriguing. Go [here](#) to create a design, answering "open" to the request in the dialog box you'll see.

The game of *SIM*. Invented by Gustavus Simmons this game is from graph theory, specifically, the easiest case of Ramsey theory. But it can be played by preschoolers as well as adults! Two players take turns coloring the edges of the complete graph K_6 , each having a different color to use. The image of K_6 we use is that of a hexagon and all its diagonals. The first to draw a triangle (with its corners on three of the hexagon's corners) is the loser. So it's a game of avoidance! Go [here](#) to play, answering "open" to the request in the dialog box you'll see. Thanks again to Neil Roza.

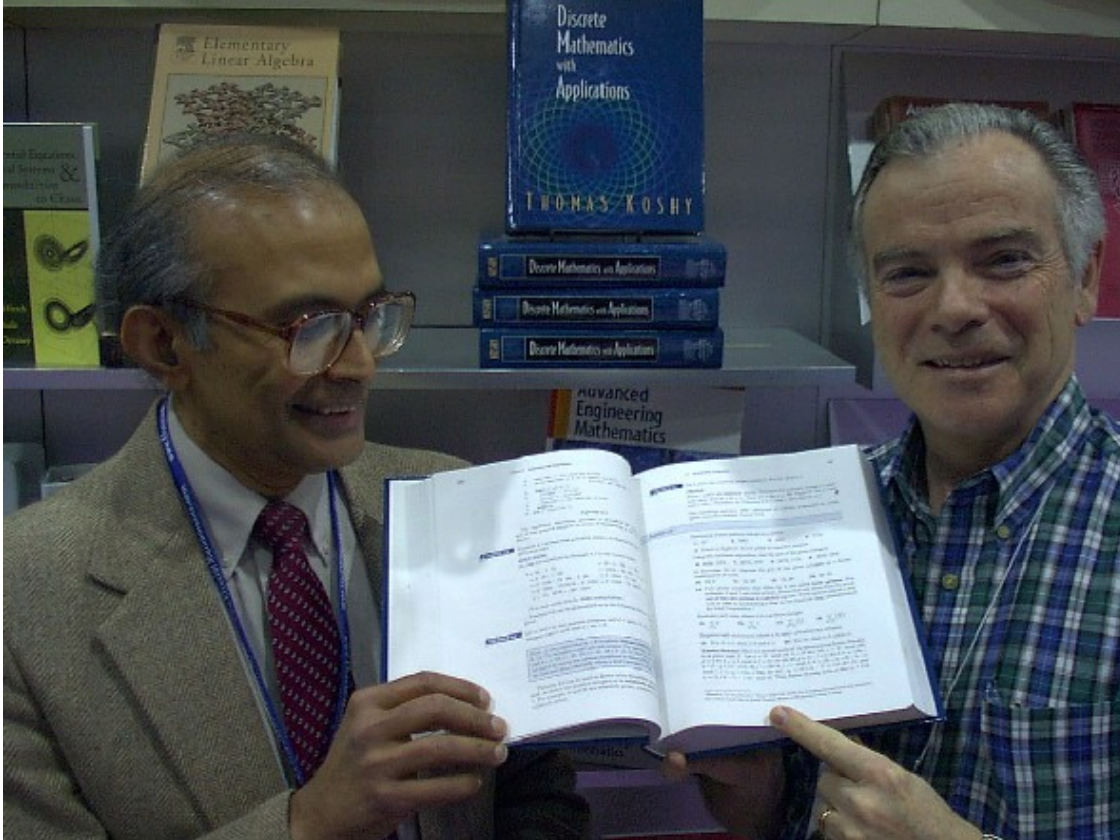
Mathematics Association of America on campus!

The Mathematics and Computer Science Department here at BSU hosted the annual November meeting of the Mathematical Association of America-Northeastern Section on Friday and Saturday, Nov. 16 and 17, 2001, and the theme was...*Recreational Mathematics*! The website for that meeting is still available here: [Fall 2001 MAA/NES meeting](#).



Bill Ritchie, founder of Binary Arts (now Thinkfun), and me. November, 2001 [Binary Arts/Thinkfun](#).

In January, 2004, I traveled to the Joint Mathematics Meeting in Phoenix where my colleague Tom Koshy from Framingham State College and I celebrated the publication of his latest textbook. Here we are pointing out a small contribution I made to this outstanding discrete mathematics text.



In January, 2006, Tom Koshy and I, and my daughter Katie, traveled to the Joint Mathematics Meetings at San Antonio, TX, where Tom and I served as facilitators at a session on Applications of Number Theory. The three of us also traveled in January, 2008 to the Joint Mathematics Meetings in San Diego. In January, 2009, Tom and I were co-facilitators at a session on The Power and Beauty of Number Theory again at the Joint Mathematics Meetings, this time in Washington, D.C. Speakers included Ken Ono, Carl Pomerance, Kirstin Eisentrager and George Andrews.

Hear ye! Hear ye!

The Mathematics Department at BSU hosted the annual meeting of the Mathematical Association of America-Northeastern Section again, on November 16-17, 2012, and the theme was *Problems, Problems, Problems: the creation, solution and publication of mathematics problems*. It was attended by over 200 mathematicians and students. The MAA described our meeting as "spectacular". The website for the meeting is available [here](#).

Awards

The Dr. V. James DiNardo Award for Excellence in Teaching .
Bridgewater State College, 1991.

Award for Distinguished College or University Teaching.
Mathematical Association of America, Northeastern Section, 1993.

Here is the full list of awardees for the Northeastern Section of the MAA, which comprises the New England states as well as the four Canadian Provinces of New Brunswick, Newfoundland, Nova Scotia, and Prince Edward Island:

1992 Frank Morgan Williams College 1993 Thomas Moore Bridgewater State College 1994 Robert Devaney Boston University 1995 Thomas Banchoff Brown University 1996 Colin Adams Williams College 1997 James J. Tattersall Providence College 1998 Robert Case Northeastern University 1999 Charles Vinsonhaler University of Connecticut 2000 Edward Burger Williams College 2001 Paul Blanchard Boston University 2002 Laura Kelleher Massachusetts Maritime Academy 2003 Emma Previato Boston University 2004 P. Joseph McKenna University of Connecticut 2005 David Abrahamson Rhode Island College 2006 Gilbert Strang Massachusetts Institute of Technology 2007 Kenneth Gross University of Vermont 2008 David Carhart Bentley College 2009 Solomon Friedberg Boston College 2010 Susan Loepp Williams College 2011 Joseph Silverman Brown University 2012 Margaret Robinson Mount Holyoke 2013 No award given

Publications

A note on the distribution of primes in arithmetic progressions. .

The Journal of Recreational Mathematics, 1972, 5(4), p. 253. This article has been referenced in the On-line Encyclopedia of Integer Sequences (OEIS) re sequence A006055.

Cayley's Color Group.

The Mathematics Teacher, 1973, 66(7), pp. 615-618.

Order as a subgroup-lattice homomorphism (with H. D'Alarcao).

American Mathematical Society Notices, 1973, 20(2), A-253.

Lagrange's Theorem Revisited (with H. D'Alarcao).

American Mathematical Monthly, 1975, 2(3), pp. 270-273.

Complete residue systems in the sequence of primes (with R. F. Sutherland).

Journal of Undergraduate Mathematics, 1975, 7(1), pp. 49-50.

Euler's formula and a game of Conway's (with H. D'Alarcao).

Journal of Recreational Mathematics, 1976, 9(4), pp. 249-251. This paper has been referenced in *The Colossal Book of Mathematics* by Martin Gardner, W. W. Norton & Company, 2001, p. 492 and in *Winning Ways for Your Mathematical Plays* by Berlekamp, Conway and Guy, Academic Press, 1982, p.568 as well as the book *Euler's Gem* by D. S. Richeson, Princeton University Press (2008), p. 298 and the article Ultimately Bipartite Games, by Cairns and Ho, in the Australasian Journal of Combinatorics, Vol. 8, 2010, pp. 213-220

Isomorphisms between groups of rational numbers.

Mathematical Gazette: the Journal of the British Mathematical Association, 1980, 64, pp. 286-287.

An aspect of group theory in residue designs. .

Two-Year College Mathematics Readings, Warren Page, editor, The Mathematical Association of America, 1981, pp. 254-257.

What hath Rubik wrought?.

Bridgewater Review, 1984, 2(2), pp. 25-26.

SIM on a microcomputer.

Journal of Recreational Mathematics, 1987, 19(1), pp. 25-29. This paper has been referenced in the book *Computers and Games*, Marsland and Frank, editors, The Proceedings of the Second International Conference on Computers and Games, CG 2000, in Hamamatsu, Japan, published by Springer, 2001

Counting bit strings with a single occurrence of 00.

Pi Mu Epsilon Journal, 1988, 8(9), pp. 572-575.

Euclid's algorithm and Lamé's theorem on a microcomputer.

The Fibonacci Quarterly, 1989, 27(4), pp. 290-295. This paper is referenced in the article *Elementary Properties of the Subtractive Euclidean Algorithm*, by A. Knopfmacher, The Fibonacci Quarterly, Vol. 30, No. 1, Feb. 1992, p. 83 and on the website [Cut-the-Knot](#)

On the least absolute remainder Euclidean algorithm.

The Fibonacci Quarterly, 1992, 30(2), pp. 161-165. This article has been referenced in "The Euclidean algorithm in algebraic number fields" by Franz Lemmermeyer, Expositiones Mathematicae, Vol. 13, No. 5 (1995) and in the book Dependence Analysis by Uta Banerjee, Kluwer Academic Publishers (1997)

Triangle Trek.

Bridgewater Magazine, 1992, 2(4), pp. 12-14.

Was Gauss Smart?

Math Horizons, November, 1999, p. 24.

This article has appeared in a book collection

The Edge of the Universe: Celebrating Ten Years of Math Horizons

Deanna Haunsperger and Stephen Kennedy, Editors, pub. by the Mathematical Association of America, 2006.

[Want to read the article?](#)



An Error Found in Lucacs and Tarjan's Mathematical Games (with J. Hatch and M. LaPlante.)
The Journal of Recreational Mathematics, 2003, 31(1), pp.25-28.

On representations of squares using triangular numbers, submitted to
Math Horizons, November, 2011

When two triangular numbers sum to a third, Mathematical Spectrum Vol. 44, Number 2, pg. 50

A note on cubes and the differences of squares, accepted by
Mathematical Spectrum, December 22, 2011

Two notes on certain relationships among triangular numbers,
Mathematical Spectrum, Vol. 45, No. 2, 2012/2013, p. 62, p. 75

Concatenation of consecutive Fibonacci and Lucas numbers: a lesson in patterns of Divisibility and Proof,
in the Ontario Mathematics Gazette, Vol. 52., No. 1, September, 2013

An investigation relating square and triangular numbers,
appeared in the Ontario Mathematics Gazette, Vol. 52, No. 2, December, 2013

The infinity of triangular oblong numbers,
accepted (7/3/2013) by and to appear in Mathematical Spectrum

Some observations on oblong numbers,
in Mathematical Medley, a publication of the Singapore Mathematical Society, Vol. 39, No. 1, 2013

Generalized Mersenne numbers and primitive Pythagorean triples,
accepted (8/7/2013) and to appear in Mathematical Medley, a publication of the Singapore Mathematical Society

A note on certain inversions in the ring of integers modulo a power of 2,
accepted (8/7/2013) and to appear in Mathematical Medley, a publication of the Singapore Mathematical Society

A note on pentagonal binomial coefficients,
submitted (3/31/2014) to the Pi Mu Epsilon Journal

Some representation problems involving figurate and Jacobsthal numbers,
submitted (11/27/2013) to Mathematical Medley, a publication of the Singapore Mathematical Society

Pythagorean triples challenge,
accepted (3/14/2014) by Girls'Angle Bulletin and appearing in Vol. 7, No. 4, April, 2014

Personalized Magic Squares,
submitted (5/14/2014) to Girls'Angle Bulletin

Original Problems Published in Mathematics Journals

(1) *Prove there are only four Mersenne numbers that are also triangular numbers*
The Journal of Recreational Mathematics, 1980, 13(3), p. 218

(2) *Prove that the congruence $32k^2 + 21k + 14 \equiv 0 \pmod{2^n}$ always has a solution, for all $n > 0$.*
Mathematics Magazine, 1981, 54(3), p. 142.

(3) *Show that the equation $(2^x - 1)(2^y - 1) = 2^{(2^z)} + 1$ is impossible in positive integers x, y and z .*
Two Year College Mathematics Journal, 1982, 13(3), p. 208.

(4) If n is an even perfect number then $n - \phi(n)$ is the square of an integer, where ϕ is Euler's phi-function.

Find infinitely many n such that $n - \phi(n)$ is a square.

Crux Mathematicorum, 1988, 14(3), p. 93.

(5) Under what conditions on the positive integers a and b will the sides of a nondegenerate triangle be formed by (i) a , b and $\gcd(a, b)$ and (ii) a , b and $\text{lcm}(a, b)$?

Pi Mu Epsilon Journal, 1990, 9(3), p. 199.<

(6) For $a < b < c$ positive integers, if $\gcd(a, b) = 1$ and $a^2 + b^2 = c^2$, then we say (a, b, c) is a primitive Pythagorean triple (PPT). If both a and c are primes we call it a prime PPT. (i) If (a, b, c) is a prime PPT, deduce that $b = c - 1$. (ii) Find all prime PPTs in which a and c are twin primes, or both are Mersenne primes, or both are Fermat primes, or one is Mersenne and the other Fermat.

Pi Mu Epsilon Journal, 1993, 9(9), p. 618

(7) Problem 926

Pi Mu Epsilon Journal, 1997, 10(7), pp. 579-580.

(8) Let $D(n)$ be the sum of the (base 10) digits of the positive integer n . Are there twin primes p and $p+2$ such that $D(p) = D(p+2)$?

Pi Mu Epsilon Journal, 2001, 11(5), p. 273.

(9) Let $t(n)$ be the n th triangular number, defined by $t(n) = t(n-1) + n$, for $n > 1$, with $t(1) = 1$.

Prove: $\gcd(t(n-1), t(n)) \cdot \gcd(t(n), t(n+1)) = t(n)$, for all $n > 1$.

Pi Mu Epsilon Journal, 2004, #1081, p. 216.

(10) The number 99

Pi Mu Epsilon Journal, 2008, 12(8), p. 494

(11) Every even perfect number is both the sum and the difference of two distinct deficient numbers.

Pi Mu Epsilon Journal, Fall, 2008, p.560

(12) In what bases are 121, 232 and 343 perfect squares? ,to appear in Crux Mathematicorum, Spring 2011

(13) Integers not the product of two deficient , School Science and Mathematics Journal, Problem 5165, May, 2011

(14) Sophie Germain and Mersenne primes , Math Horizons, a publication of the MAA, Problem 293, Sep, 2011, p. 30

(15) Cubes and even perfect numbers , Problem 5209, in School Science and Mathematics Journal, Apr., 2012

(16) Even perfect numbers as fourth powers ,submitted to School Science and Mathematics Journal

(17) The primality of 9^{n+2} and 9^{n-2} , Pi Mu Epsilon Journal (problem 1254, Spring/2012)

(18) Certain simultaneously composite rotation numbers , The Pentagon, problem 700, Vol. 71, No. 2, pg. 41

(19) Certain exponential trinomials as perfect squares , The Pentagon, problem 699, Vol. 71, No. 2, pg. 41

(20) Triangular numbers as sums of consecutive powers of 2 , Problem 44.6, Mathematical Spectrum, Vol. 44, No. 3

(21) On a certain diophantine equation , accepted by and to appear in Pi Mu Epsilon Journal

(22) On the compositeness of an exponential expression , accepted by and to appear in Pi Mu Epsilon Journal

(23) Fourth powers and triangular numbers , accepted by and to appear in School Science and Mathematics Journal

(24) Show that a certain automatically generated sequence consists solely of triangular numbers , Problem 690, Vol. 71, No. 1, The Pentagon

(25) On the number of idempotents in a ring with unity , Mathematics Magazine, December, 2012, p.381, as Q1026, solution in Vol. 86, No. 1, Feb., 2013, p. 71

(26) Expressing triangular numbers using a certain number of pentagonal numbers , Problem 689, Vol. 71, No. 1, The Pentagon

(27) *On pentagonal numbers not the sum of a certain number of squares* , accepted by and to appear in Pi Mu Epsilon Journal

(28) *On cubes the sum of two triangular numbers* , submitted to Parabola, a publication of the University of New South Wales, AU, November, 2011

(29) *Triangular numbers as sums of consecutive integers* , submitted to Math Horizons, a publication of the MAA, November, 2011

(30) *On cubes that are the sum of Mersenne numbers* , Problem 44.11, Mathematical Spectrum, Vol. 44, No. 3, p. 139

(31) *Find infinitely many squares of triangular numbers each the sum of a square and a cube* , Problem 714 in The Pentagon, Fall, 2012

(32) *On sums of two squares and two cubes* , Problem 711 in The Pentagon, Fall, 2012

(33) *On squares expressible in a certain binomial form* , accepted by and to appear in Pi Mu Epsilon Journal

(34) *On squares expressible in a certain trinomial form* , Problem 5225 in School Science and Mathematics Journal, Nov. 2012

(35) *On an algorithmic description of sums of three squares among the integer squares* , accepted by The Pentagon (12/15/2011)

(36) *On pentagonal numbers and the differences of squares* , Math Horizons, problem 274, April, 2012, p. 30.

(37) *Bases with at least two repdigits decimally triangular* , accepted (12/22/2011) by and to appear in The Pentagon

(38) *A base in which certain repdigits are triangular numbers* , Problem 5238, in School Science and Mathematics Journal, Jan., 2013 (12/23/2011)

(39) *Fibonacci numbers as differences of hexagonals* , submitted to Mathematics Magazine (1/13/2012)

(40) *Pentagonal numbers as the difference of hexagonals and vice versa* , submitted to The Pentagon (1/14/2012)

(41) *On the gcds of triangular and pentagonal numbers of the same rank* , Problem 5220 in School Science and Mathematics Journal, Oct. 2012, Vol. 112, NO. 6, page 392

(42) *An observation on the gcds of powers of 6 and hexagonal numbers* , accepted (2/4/2012) by and to appear in the Pi Mu Epsilon Journal

(43) *On which Fibonacci numbers are the values of gcds of certain polygonal numbers* , Problem 5244, in School Science and Mathematics Journal, Feb., 2013

(44) *Powers of 2 and the differences of pentagonal numbers* , accepted (12/19/2013) by The Pentagon

(45) *A question on sums of three cubes* , Math Horizons, February, 2013, p. 31, solution September 2013, p.31

(46) *A certain representation of the squares of Fibonacci numbers* , submitted (2/17/2012) to The Fibonacci Quarterly

(47) *A congruence inspired by Fermat's little theorem* , Problem 5260 in School Science and Mathematics Journal , May, 2013

(48) *A congruence involving the Euler phi-function* , The Pentagon, problem 701, Vol. 71, No. 2, pg. 42

(49) *On Pythagorean triples with an abundant and a deficient leg* , submitted(5/30/12) to The Pentagon

(50) *Representations of integers using arithmetic functions* , School Science and Mathematics Journal, Vol. 113, No. 4, pagh 211

(51) *On divisors of the products of certain arithmetic functions* , submitted (6/27/12) to The Pentagon

(52) *Cubes among sums of certain figurate numbers* , in Mathematical Spectrum, problem 45.11, Vol. 45, No. 3., p. 132

(53) *Certain prime powers as the sum of two squares* , Problem 712 in The Pentagon, Fall, 2012

- (54) *Powers of 2 represented as a sum of two figurate numbers* , Problem 713 in The Pentagon, Fall, 2012
- (55) *Pentagonal numbers as the power of a single prime* , Problem 691, Vol. 71, No. 1, The Pentagon
- (56) *Triangular numbers that are twice a prime* , appeared as Problem 5278 in the School Science and Mathematics Journal, December, 2013
- (57) *Certain numbers as a sum of two composites* , Problem 725 in The Pentagon, Spring, 2013, p. 22
- (58) *Primes represented by an exponential form* , Problem 722 in The Pentagon, Spring, 2013, p. 21
- (59) *A sequence of divisions performed on a special exponential form* , appeared in the School Science and Mathematics Journal as Problem 5284, January, 2014
- (60) *Triangular numbers and certain quadratic forms* , accepted (2/6/2013) by Mathematical Spectrum
- (61) *On a prime generating polynomial* , the Ramanujan Mathematical Society Newsletter Vol. 23, No. 4, March, 2013, page 280
- (62) *A variation on the lo shu magic square*, in the Ramanujan Mathematical Society Newsletter Vol. 23, No. 4, March, 2013, page 280
- (63) *When multiplying the digits of a number N gives N* , submitted (2/19/2013) to The Pentagon
- (64) *When do $a^n + b^n$ and $a^2 + b^3$ yield squares?*, Problem 5249, in School Science and Mathematics Journal, Mar., 2013
- (65) *Squares via a certain cubic form* , in the Ramanujan Mathematical Society Newsletter, Vol. 23, No. 4, March, 2013, page 280
- (66) *Primitive Pythagorean Triangles with one side pentagonal* , Problem 723 in The Pentagon, Spring, 2013, p. 21
- (67) *On fields with additive and multiplicative inverses the same* , Mathematics Magazine, April, 2013, p.???, as Q1029
- (68) *Pentagonal numbers expressible as sums of three Jacobsthal numbers* , accepted (3/4/2013) and to appear in School Science and Mathematics Journal
- (69) *A ratio of products of triangular numbers always an integer* , Problem 724 in The Pentagon, Spring, 2013, p. 21
- (70) *On primes omitted by a Legendre polynomial* , submitted (4/4/2013) to The Irish Undergraduate Mathematics Magazine
- (71) *On primes omitted by an Euler polynomial* , submitted (4/4/2013) to the Pi Mu Epsilon Journal
- (72) *Linear combinations of Jacobsthal numbers* , submitted (4/8/2013) to Eureka: the Journal of the Archimedeans
- (73) *Pythagorean triples and Jacobsthal numbers* , appeared as Problem 5272 in School Science and Mathematics Journal, November, 2013
- (74) *On products of consecutive Jacobsthal numbers* , accepted (4/25/2013) and to appear in the Pi Mu Epsilon Journal
- (75) *On representing triangular numbers using triangular numbers in a special form* , Bulletin of the Irish Mathematical Society, No. 71, summer, 2013, p. 77
- (76) *On squares represented by a sum of products of Jacobsthal numbers* , Bulletin of the Irish Mathematical Society, No. 71, summer, 2013, p. 77
- (77) *Triangular numbers as a sum of products of Jacobsthal numbers* , withdrawn (4/29/2013) from Euclides (the journal of the Dutch Association of Mathematics Teachers)
- (78) *On squares represented by an expression of pentagonal numbers* , accepted (5/16/2013) by The Mathematical Gazette (the Journal of the Mathematics Association of Great Britain) and to appear in 2015
- (79) *On two forms n^2+p having at least so many divisors* , appeared in Math Horizons, an MAA journal September, 2013, p. 30

(80) Primitive Pythagorean Triples and Mersenne numbers , submitted (5/13/2013) to The Irish Undergraduate Mathematics Magazine

(81) Divisibility of concatenated triangular numbers , accepted (5/18/2013) by Mathematical Spectrum

(82) On squares represented by two special quadratic forms , submitted (5/28/2013) to Pi in the Sky (Journal of the Pacific Institute for the Mathematical Sciences)

(83) Triangular numbers that are the sum of two squares , accepted (6/5/2013) and to appear in Math Problems Journal (University of Prishtina, Kosovo)

(84) On the equation $P(a)P(b)=2T(c)$ for certain figurate numbers , accepted (6/2/2013) by and to appear in The Pentagon

(85) A sequence of diophantine equations inspired by FLT , appeared in the Ramanujan Mathematical Society Newsletter , Volume 24, No. 1, p. 14

(86) On odd multiples of 5 occuring in primitive Pythagorean triples , submitted (6/10/2013) to The Pentagon

(87) Divisibility properties of triangular oblong numbers , accepted (7/2/2013) by Mathematical Spectrum

(88) On numbers that are the leg and hypotenuse of primitive Pythagorean triangles , appeared in the MathProblems Journal (University of Prishtina, Kosovo), Vol. 3, No. 3 (November, 2013), problem 12, p. 193.

(89) On squares that are the legs of a Pythagorean triangle , accepted (7/15/2013) by The Pentagon

(90) A certain expression and divisibility by 13 , accepted (7/23/2013) by School Science and Mathematics Journal

(91) An inequality on the central binomial coefficient , accepted (7/27/2013) by Mathematical Spectrum

(92) On two odd prime nondivisors of a special form , accepted (9/4/2013) by and to appear in The Pentagon

(93) Triangular and pentagonal numbers adding to 2^n or n^2+n^3 , accepted (9/11/2013) by and to appear in the Irish Mathematical Society Bulletin

(94) A quadratic polynomial with all triangular values for special inputs , appeared in the Ramanujan Mathematical Society Newsletter, vol. 24, no. 2, Sep. 2013, p. 37

(95) On two cubic diophantine equations , accepted (9/30/2013) by and to appear in The Pentagon

(96) On certain binomial coefficients the sum of two squares , submitted (10/25/2013) to the Ramanujan Mathematical Society Newsletter

(97) Three relations among figurate numbers , accepted (11/1/2013) and to appear in School Science and Mathematics Journal

(98) Even perfect numbers related to sums of certain figurate numbers , appeared in the Irish Mathematical Society Bulletin, Number 72, Winter, 2013, problem 72.3, pp. 101-102

(99) Powers of 2 as a difference of pentagonal numbers, accepted (12/19/2013) by and to appear in The Pentagon

(100) Divisibility of sums of squares and the product of their bases, accepted (1/6/2014) by the Pi Mu Epsilon Journal

(101) Divisibility tests in octal notation, submitted (1/24/2014) to the MathAMATYC Educator

(102) Certain properties of star numbers, accepted (1/27/2014) by and to appear in The Pentagon

(103) Wrongly remembered even perfect form classified as deficient or abundant , appeared in the School Science and Mathematics Journal as Problem 5290, February, 2014

(104) Infinitely many squares within two simple quadratic forms, submitted (1/29/2014) to The Octogon Mathematical Magazine (Romania)

(105) Equal sums and differences of oblong numbers, accepted (2/4/2014) by and to appear in The Pentagon

- (106) *On the diophantine equation $a^2+b^2-c^2=c-a-b$* , submitted (2/12/2014) to the MathAMATYC Educator
- (107) *Arithmetic of hexagonal numbers* , submitted (2/13/2014) to School Science and Mathematics Journal
- (108) *On the highest power of 2 or 3 dividing both legs of a Pythagorean Triangle* , appeared in Girls'Angle Bulletin, Vol. 7, No. 4, April, 2014, p. 22
- (109) $\tau(\tau(2015n))$ and $\phi(\tau(2015n))$, accepted (3/5/2014) by and to appear in The Pentagon
- (110) *Belpheger's prime number inspiration* , submitted (3/6/2014) to and accepted (3/8/2014) by and to appear in School Science and Mathematics Journal
- (111) *On squares represented by $a^2+b^2+3c^2$* , accepted (3/8/2014) by the MathAMATYC Educator
- (112) $n\phi(n)$ is the fourth power of an integer infinitely often , appeared in School Science and Mathematics Journal, Problem 5302, April, 2014
- (113) *Primitive Pythagorean triples (a,b,c) with abc divisible by 120* , submitted (4/3/2014) to The Pentagon
- (114) *Primitive Pythagorean triples (a,b,c) with $a+b+c$ divisible by 60* , submitted (4/5/2014) to School Science and Mathematics Journal

Letters to editors and authors

- (1) *An extension of a discovery of Goormaghtigh*
in the Journal of Recreational Mathematics, Vol. No. 72, pp. 101-102
- (2) *On the magic hexagon of Clifford Adams*
in Mathematical Gems by Ross Honsberger (Dolciani series of the MAA), page ??
- (3) *On repdigit numbers that are the sum of a square, a triangular number and a pentagonal number*
appeared in Mathematical Spectrum, Vol. 46, N0. 3, pp. 137-138
- (4) *On numbers the sum of two squares and two cubes*
submitted (6/6/2013) to Mathematical Spectrum